## 1. Convert color to Grayscale and Negative Transformation

```
% Read the color image
img = imread('pic1.jpg');
% Convert to grayscale
grayImg = rgb2gray(img);
% Negative transformation
Negative = imcomplement(img);
% Display the results
figure; subplot(1,3,1);
imshow(img);
 title('Original Image');
subplot(1,3,2);
imshow(grayImg);
title('Grayscale Image');
subplot(1,3,3);
imshow(Negative);
title('Negative Image');
```

## 2. Smoothing and Sharpening of the Images

```
clc;
 clear;
 close all;
 % Read the 8-bit color image
img = imread('sunflower.jpg');
img = imresize(img, 0.8);
% Display Original Image
figure('Position', [100 100 800 600]);
subplot(2,2,1);
imshow(img);
title('Original Image');
 % Image Smoothing using Average Filter
avgFilter = fspecial('average', [5 5]);
smooth_avg = imfilter(img, avgFilter, 'replicate');
subplot(2,2,2);
 imshow(smooth_avg);
title('Average Filtered Image');
% Image Sharpening
 sharp_img = imsharpen(img, ...
'Radius', 2, …
 'Amount', 1.5);
subplot(2,2,3);
 imshow(sharp_img);
 title('Sharpened Image');
```

## 3. Histogram Equalization of a Color Image

```
clc;
clear;
 close all;
 % Read the color image
img = imread('rose.webp');  % Replace with your image file
% Display original image
figure;
subplot(2,2,1);
imshow(img);
title('Original Color Image');
% Convert RGB image to HSV
hsvImg = rgb2hsv(img);
% Extract the Value (Brightness) channel
V = hsvImg(:,:,3);
 % Perform Histogram Equalization on Value channel
Veq = histeq(V);
% Replace the Value channel with equalized one
hsvImg(:,:,3) = Veq;
% Convert HSV back to RGB
equalizedImg = hsv2rgb(hsvImg);
% Display equalized color image
subplot(2,2,2);
imshow(equalizedImg);
title('Histogram Equalized Color Image');
% Display histogram of original Value channel
subplot(2,2,3);
imhist(V);
title('Original Value Histogram');
% Display histogram of equalized Value channel
subplot(2,2,4);
imhist(Veq);
```

```
title('Equalized Value Histogram');
```

## 4.Edge detection using Sobel gradient operator.

```
clc;
clear;
close all;
 % Read input image
I = imread('rose.jpg');
% Convert RGB image to grayscale
 if size(I,3) == 3
Igray = rgb2gray(I);
else
Igray = I;
end
% Convert image to double
Igray = im2double(Igray);
% Sobel gradient masks
Gx = [-1 0 1;
-2 0 2;
-1 0 1];
Gy = [-1 -2 -1;
0 0 0;
1 2 1];
 % Calculate gradients
Ix = imfilter(Igray, Gx, 'replicate');
 Iy = imfilter(Igray, Gy, 'replicate');
% Calculate gradient magnitude
G = sqrt(Ix.^2 + Iy.^2);
% Normalize gradient
 G = mat2gray(G);
% Sharpen image using gradient
```

```
alpha = 1.0;
Sharpened = Igray + alpha * G;
% Limit intensity values to [0,1]
Sharpened = min(max(Sharpened,0),1);
% Display results
figure;
subplot(2,2,1);
 imshow(Igray);
title('Original Grayscale Image');
subplot(2,2,2);
imshow(Ix,[]);
title('Gradient in X Direction');
subplot(2,2,3);
imshow(G,[]);
title('Gradient Magnitude');
subplot(2,2,4);
 imshow(Sharpened);
title('Sharpened Image');
```

## 5.Erosion and dilation operations

```
clc;
 clear;
close all;
 f = imread('grass.jpg');
se = strel('square', 20);
fo = imopen(f, se);
figure, imshow(f),title('Input image');
figure, imshow(fo),title('Opening of input image');
fc = imclose(f, se);
figure,imshow(fc),title('Opening of input image');
foc = imclose(fo, se);
```

```
figure, imshow(foc),title('Closing of opened image');
```

## 6. Image segmentation

```
clc;
clear;
close all;
img = imread('fig.jpg');
gray = rgb2gray(img);
threshold = 20;
mask = gray > threshold;
mask = bwareaopen(mask,500);
 boundary = bwperim(mask);
marked_img = img;
R = marked_img(:,:,1);
G = marked_img(:,:,2);
 B=marked_img(:,:,3);
R(boundary) = 0;
G(boundary) = 255;
B(boundary) = 0;
marked_img(:,:,1) = R;
 marked_img(:,:,2) = G;
 marked_img(:,:,3) = B;
subplot(1,3,1);
imshow(img);
title('Original Image');
subplot(1,3,2);
imshow(mask);
title('Segmented Region');
subplot(1,3,3);
imshow(marked_img);
```

```
title('Segmented Image with Boundary');
```

## 7. Convert an RGB image into an HSV and split the channels

```
rgbImage = imread('img.jpg');  % Replace with your image file
hsvImage = rgb2hsv(rgbImage);
% Display the original RGB image
figure;
subplot(2,2,1);
imshow(rgbImage);
title('RGB Image');
% Display the HSV image
subplot(2,2,2);
imshow(hsvImage);
title('HSV Image');
% Display the Hue component
subplot(2,2,3);
imshow(hsvImage(:,:,1));
title('Hue (H)');
% Display the Saturation component
subplot(2,2,4);
imshow(hsvImage(:,:,2));
title('Saturation (S)');
% Display the Value component
figure;
imshow(hsvImage(:,:,3));
title('Value (V)');
```